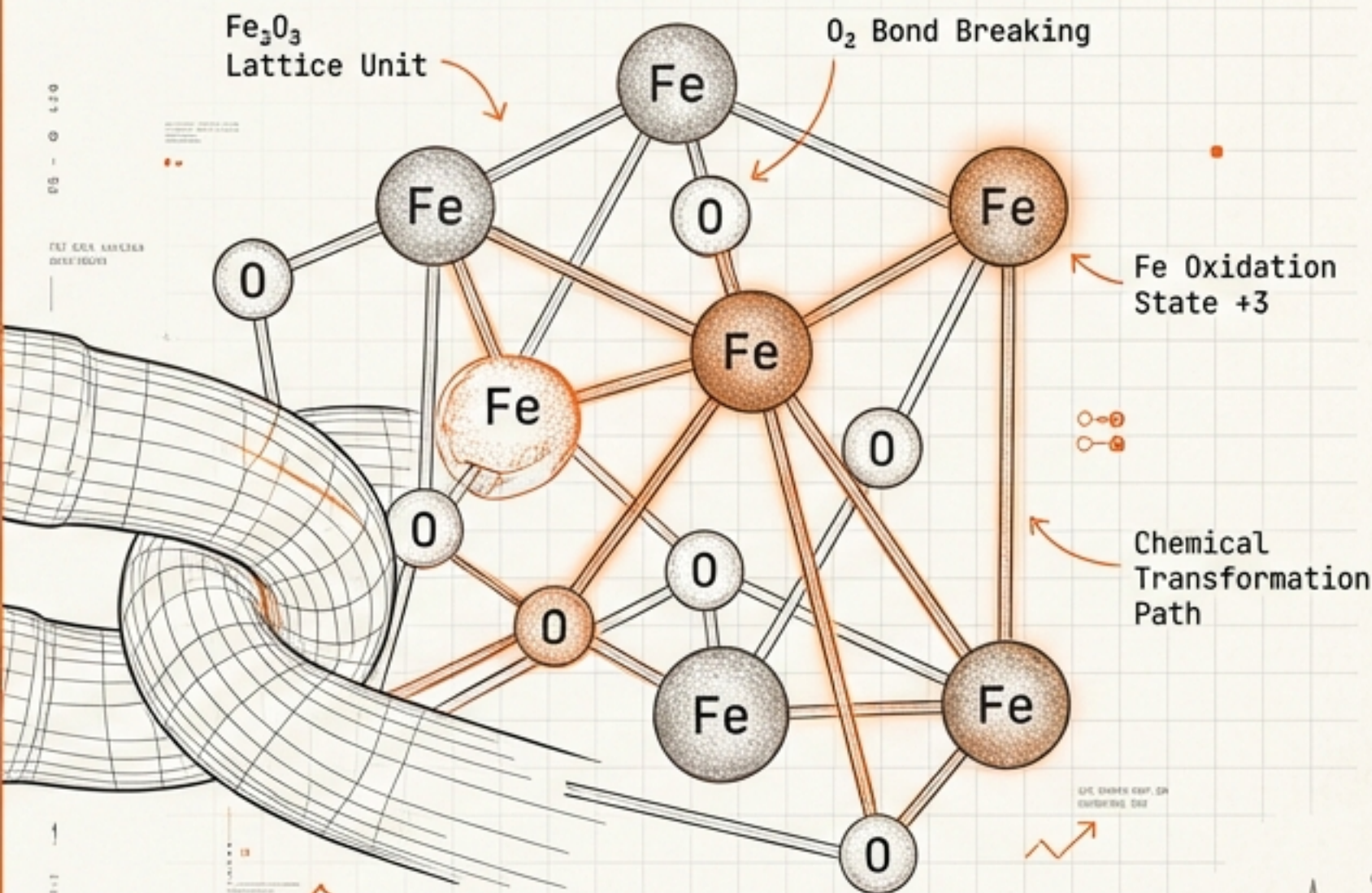
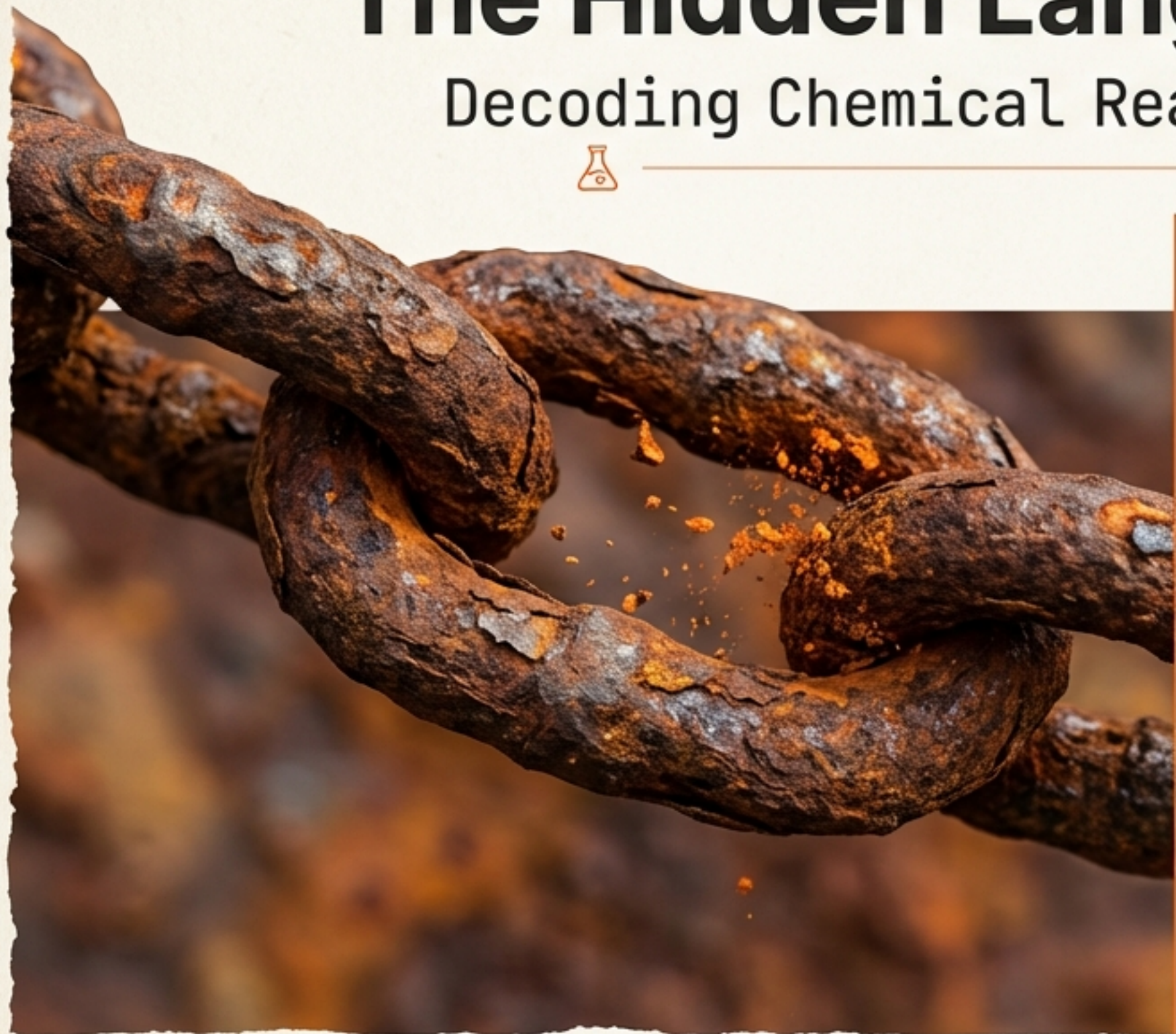
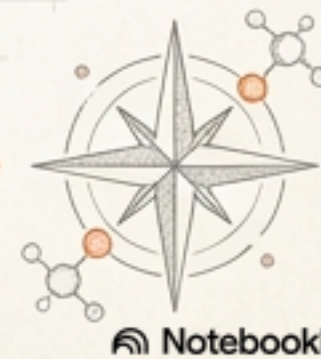


The Hidden Language of Matter

Decoding Chemical Reactions and Equations

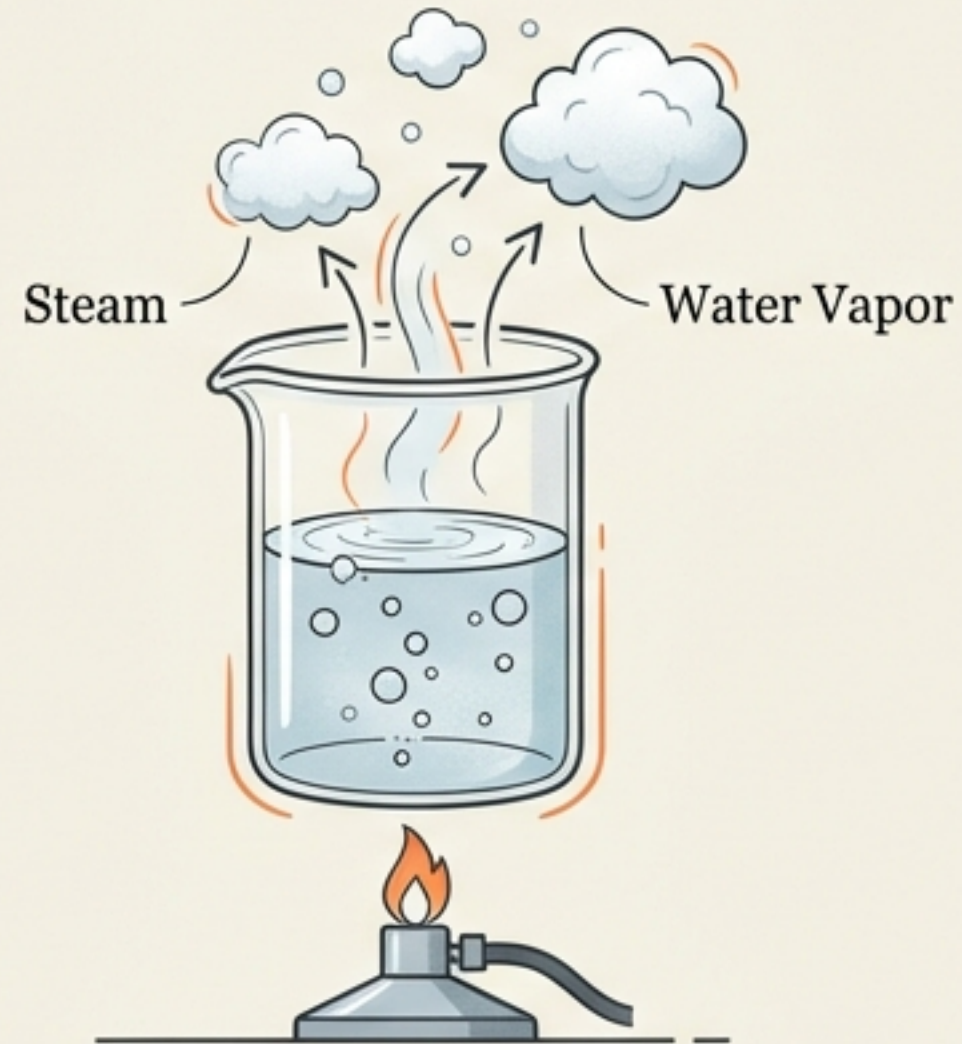


Every second, the universe rearranges itself. From the rusting of iron to the digestion of food, matter is constantly changing identity. To understand these changes, we must learn to read the equation.



Change is the Only Constant

Physical Change



Shape or State changes.
Substance remains the same.

Chemical Change



1. Milk curdling in a glass.



2. A rusted iron pan.



3. Fermenting grapes.



4. Food cooking in a pot.

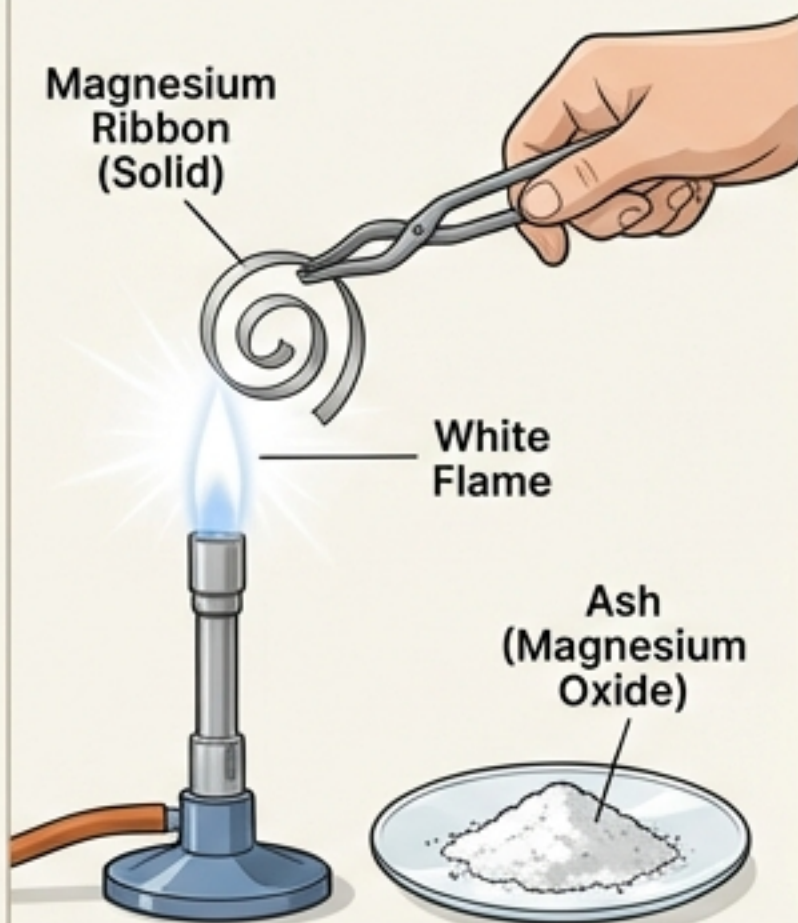
Identity changes.
A new substance is formed.

Key Insight: Whenever a chemical change occurs, we say a Chemical Reaction has taken place.

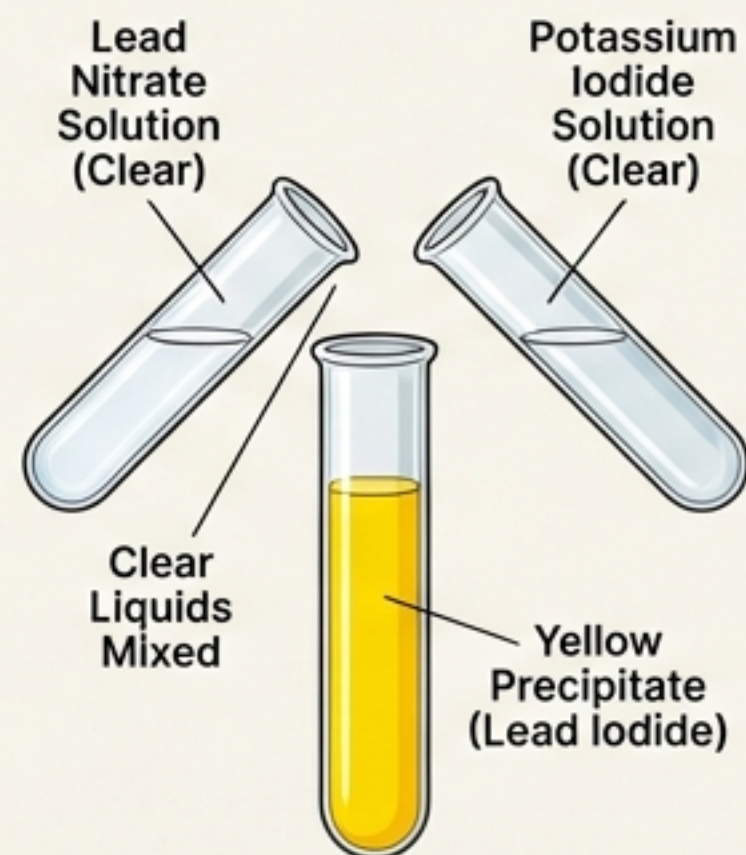
The Detective's Toolkit

Observable indicators that a chemical reaction has occurred.

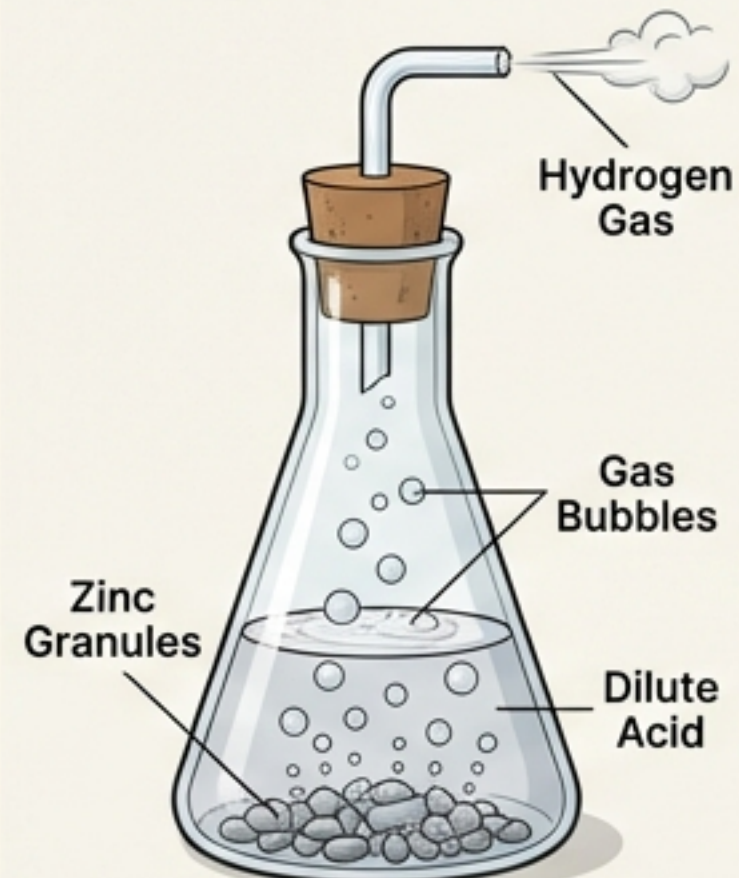
Change in State



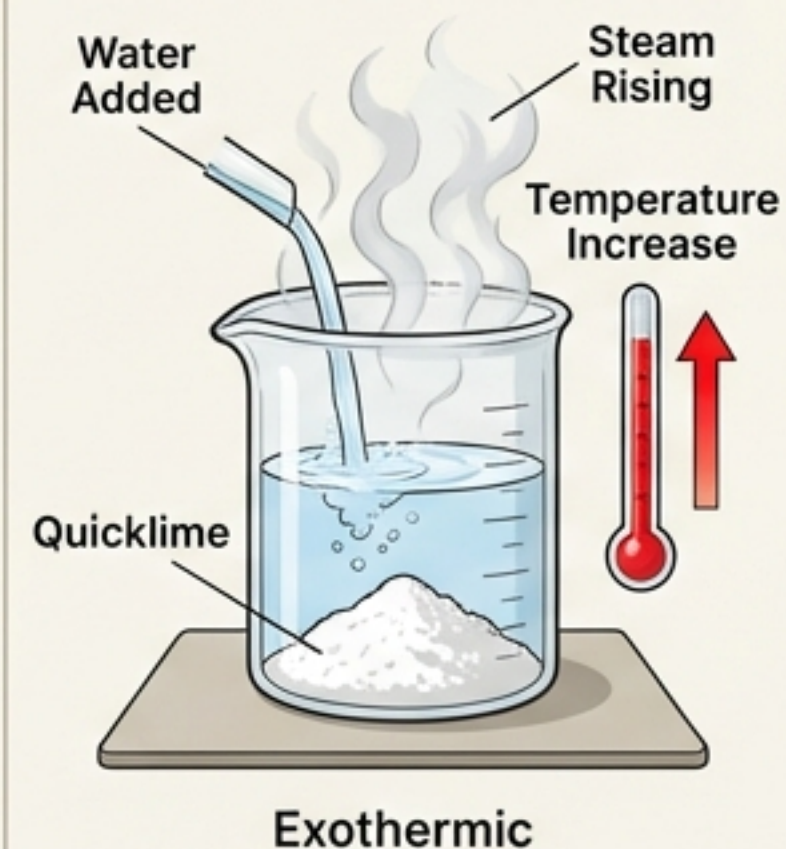
Change in Colour



Evolution of Gas



Change in Temperature



Caution: Handling acids and burning magnesium requires protective eyewear and teacher assistance.

Cracking the Code

Describing reactions in sentences is inefficient. We use chemical equations to capture the transformation concisely.

Sentence: When a magnesium ribbon is burnt in oxygen, it gets converted to magnesium oxide.



Word Equation: Magnesium + Oxygen → Magnesium Oxide



Direction of Reaction

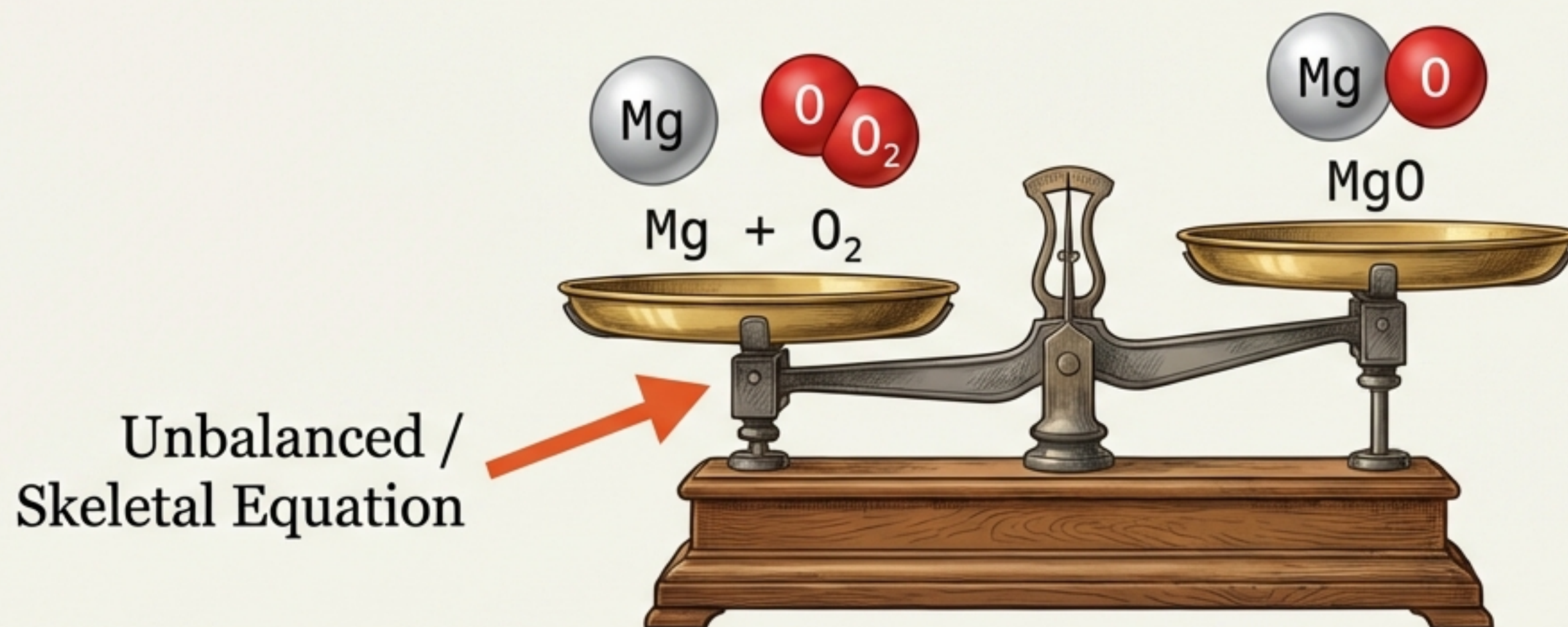
Skeletal Equation: $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$

Reactants (LHS)

Products (RHS)

The Law of Conservation of Mass

Mass can neither be created nor destroyed in a chemical reaction.

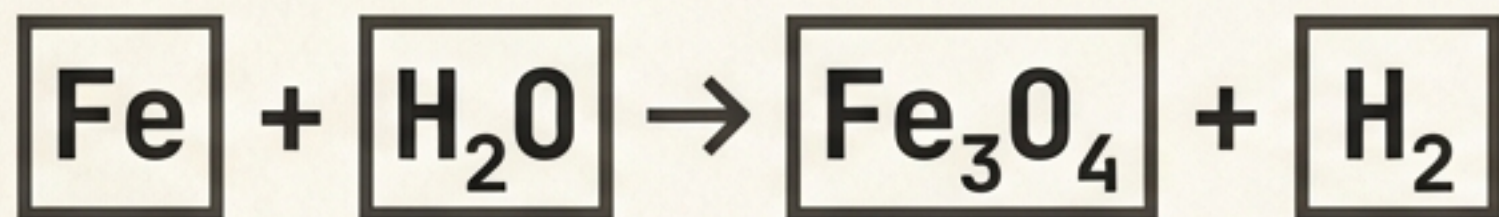


Element	LHS	RHS
Mg	1	1
O	2	1

Master Class: Balancing Equations in Charcoal

The 'Hit-and-Trial' Method: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$

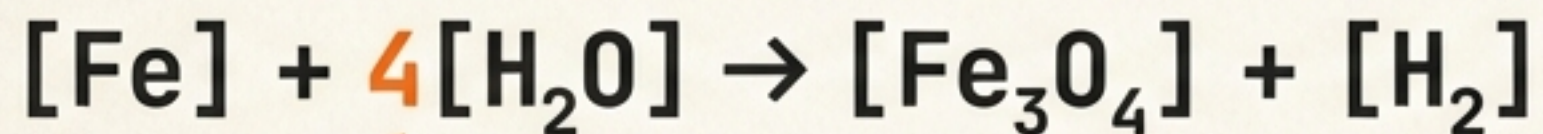
1. Box It



2. Count

Element	LHS (Reactants)	RHS (Products)
Fe	1	3
H	2	2
O	1	4

3. Equalise

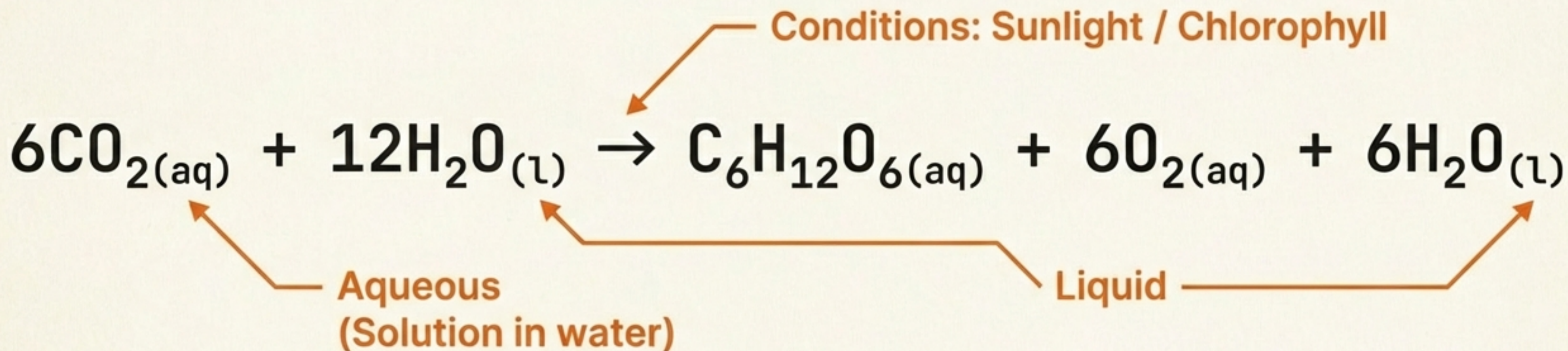


4. Finalize



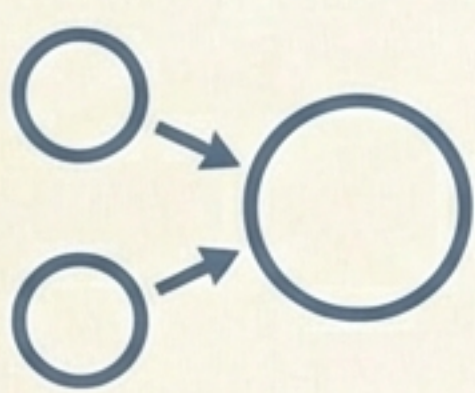
Refining the Code

To make equations informative, we add physical states and necessary conditions.

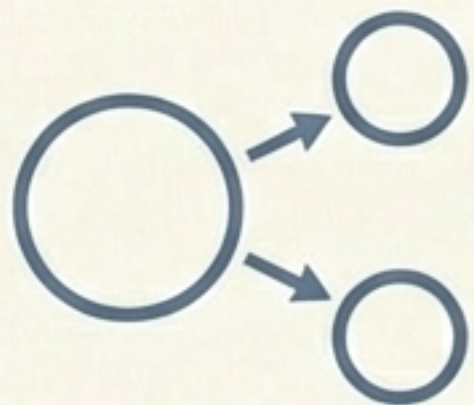


The Patterns of Interaction

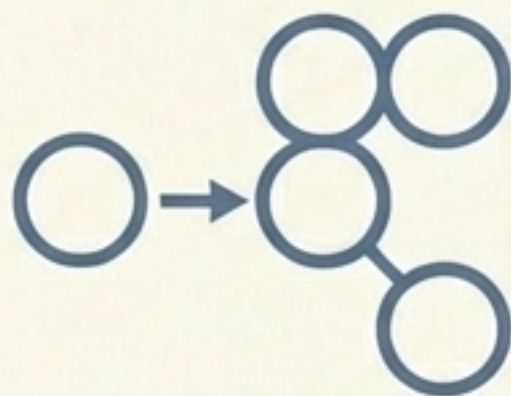
We classify chemical changes into 5 archetypes.



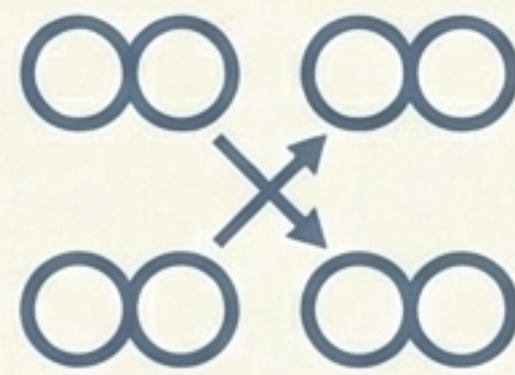
Combination
The Union



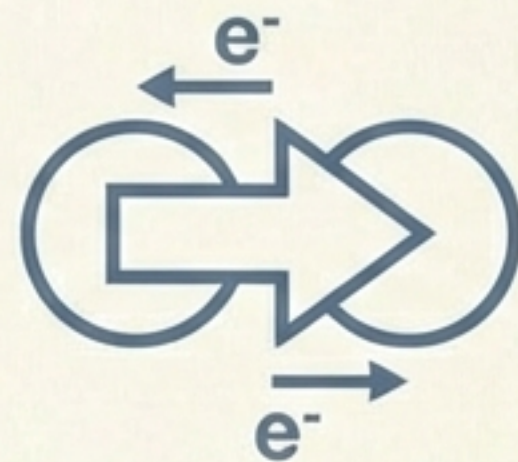
Decomposition
The Breakup



Displacement
The Substitute



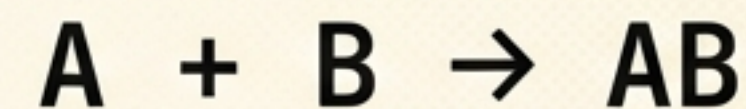
Double
Displacement
The Exchange



Redox
The Tug-of-War

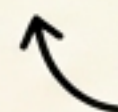
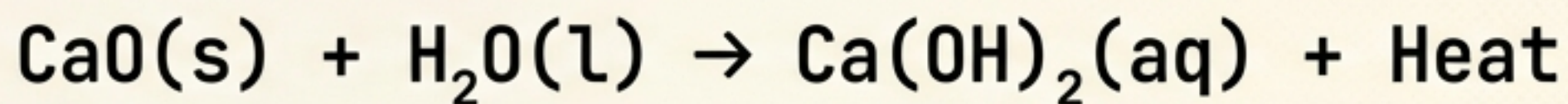


Pattern 1: Combination (The Union)

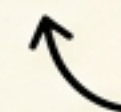


Definition:

A single product is formed from two or more reactants.



Quick Lime



Slaked Lime

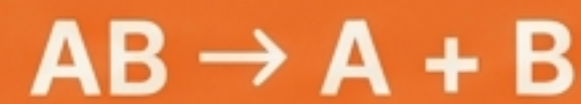


Quick Lime Reaction

Exothermic Reactions:

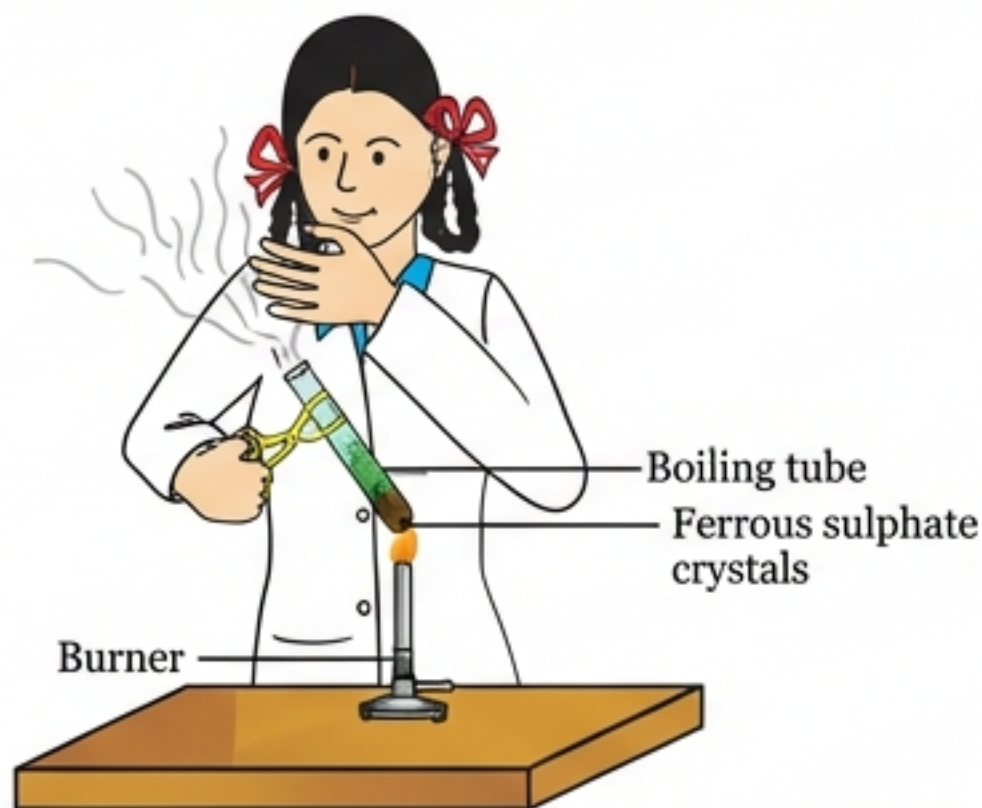
Reactions that release heat. Examples include burning natural gas and Respiration.

Pattern 2: Decomposition (The Breakup)

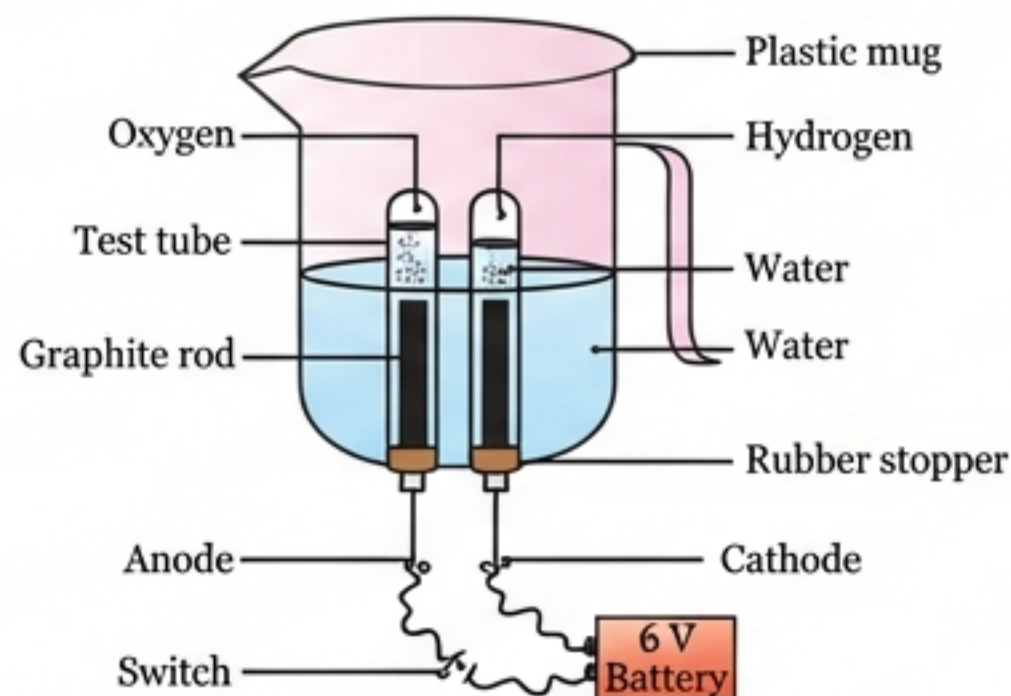


A single reactant breaks down into simpler products. This requires energy.

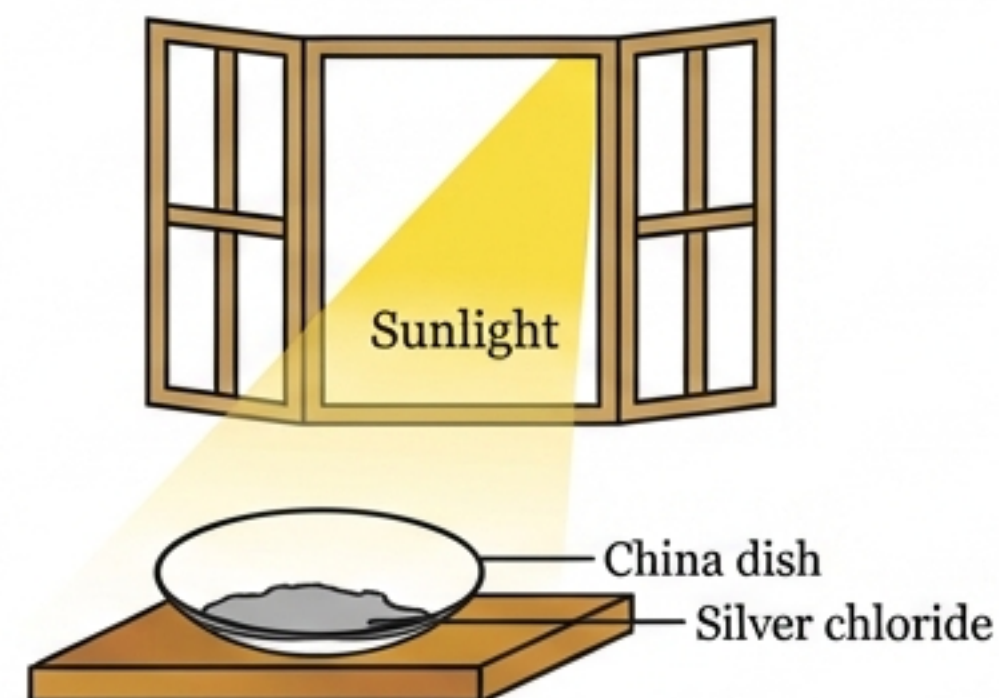
Thermal (Heat)



Electrical



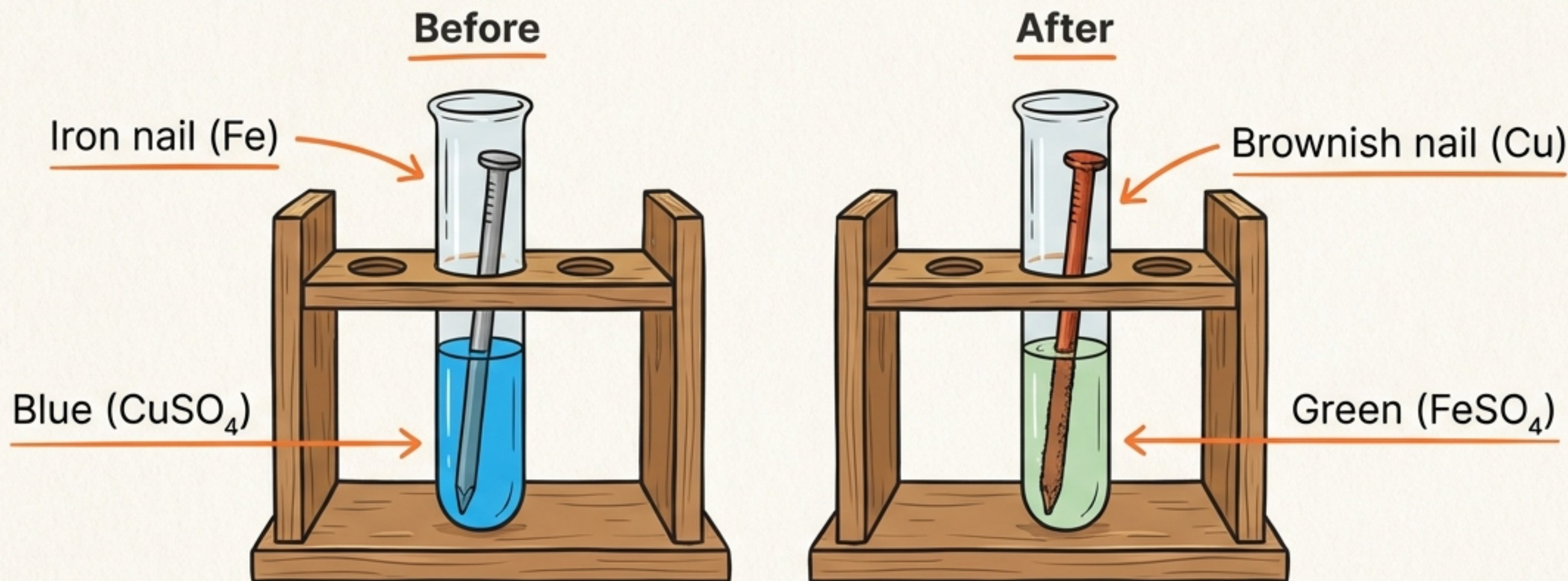
Photolytic (Sunlight)



These are Endothermic Reactions (Energy is absorbed).

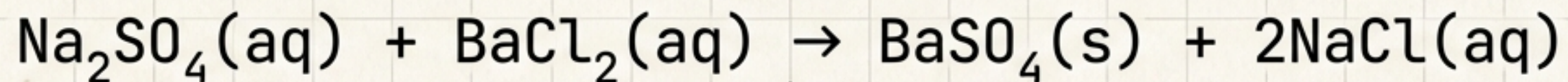
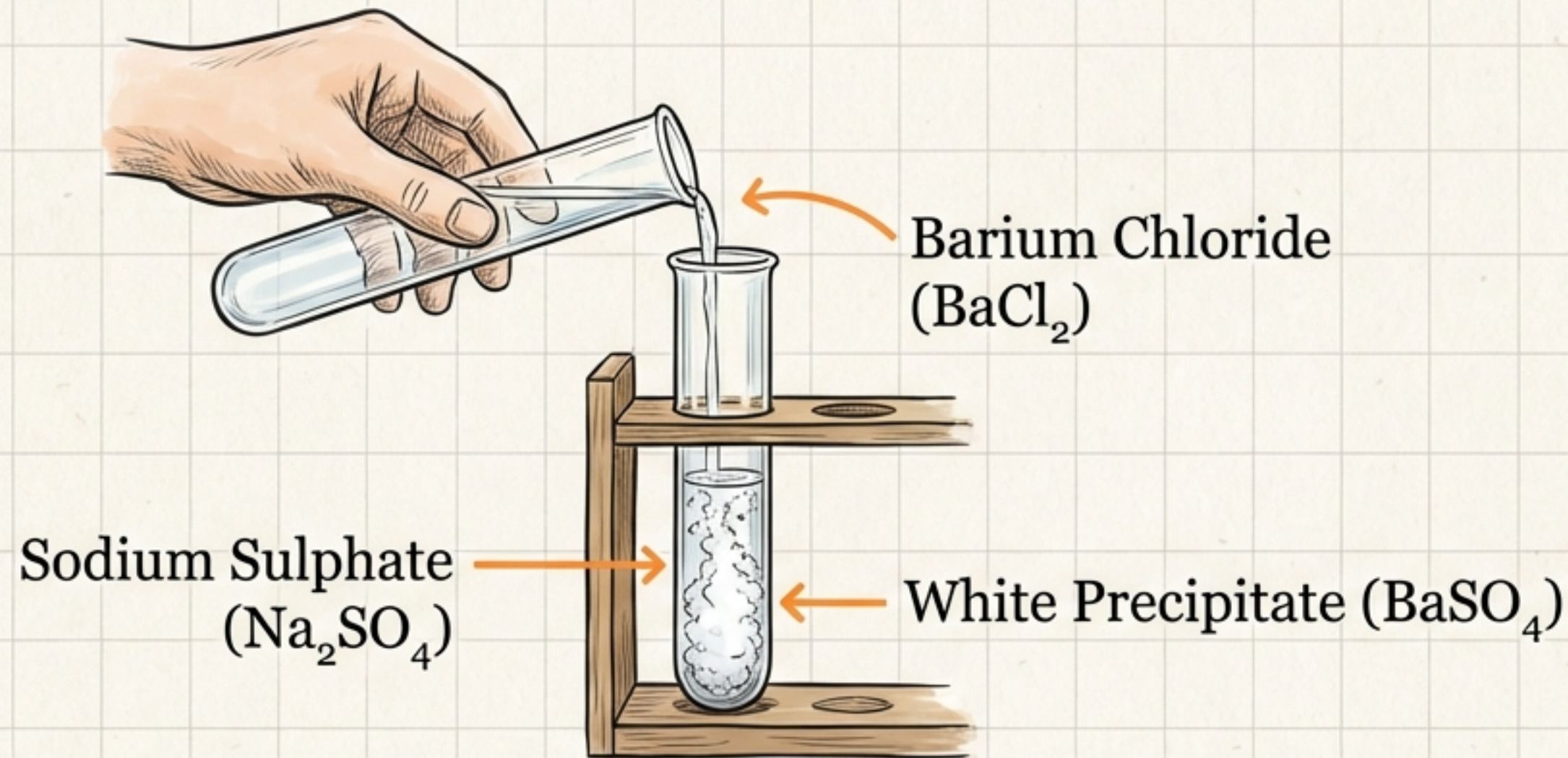
Pattern 3: Displacement (The Substitute)

Charcoal JetBrains Mono
 $A + BC \rightarrow AC + B$



Iron is more reactive than Copper, so it displaces it.

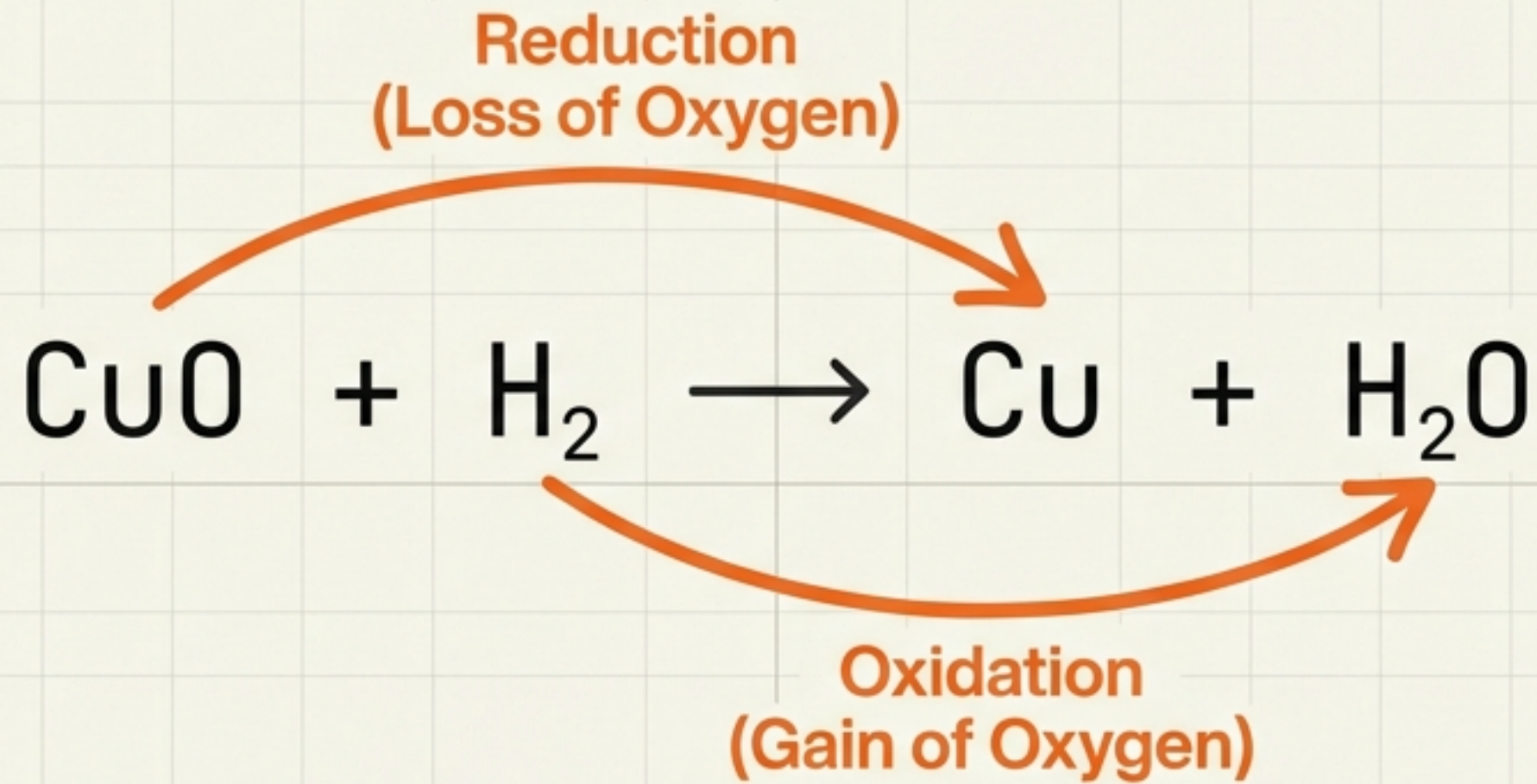
Pattern 4: Double Displacement (The Exchange) $AB + CD \rightarrow AD + CB$



Precipitation Reaction: Any reaction that produces an insoluble solid (precipitate).

Pattern 5: Oxidation and Reduction (Redox)

The Tug-of-War for Oxygen.

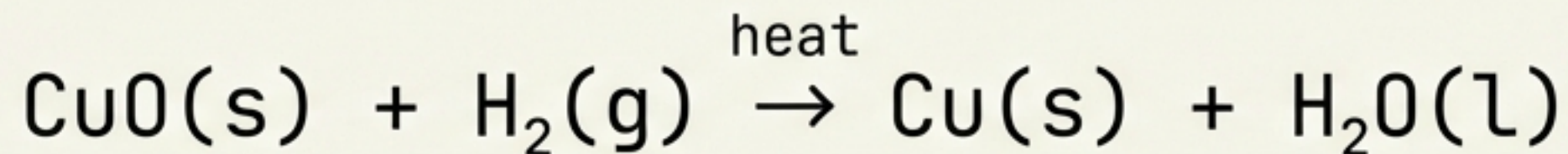


Oxidation

Gaining Oxygen or Losing Hydrogen.

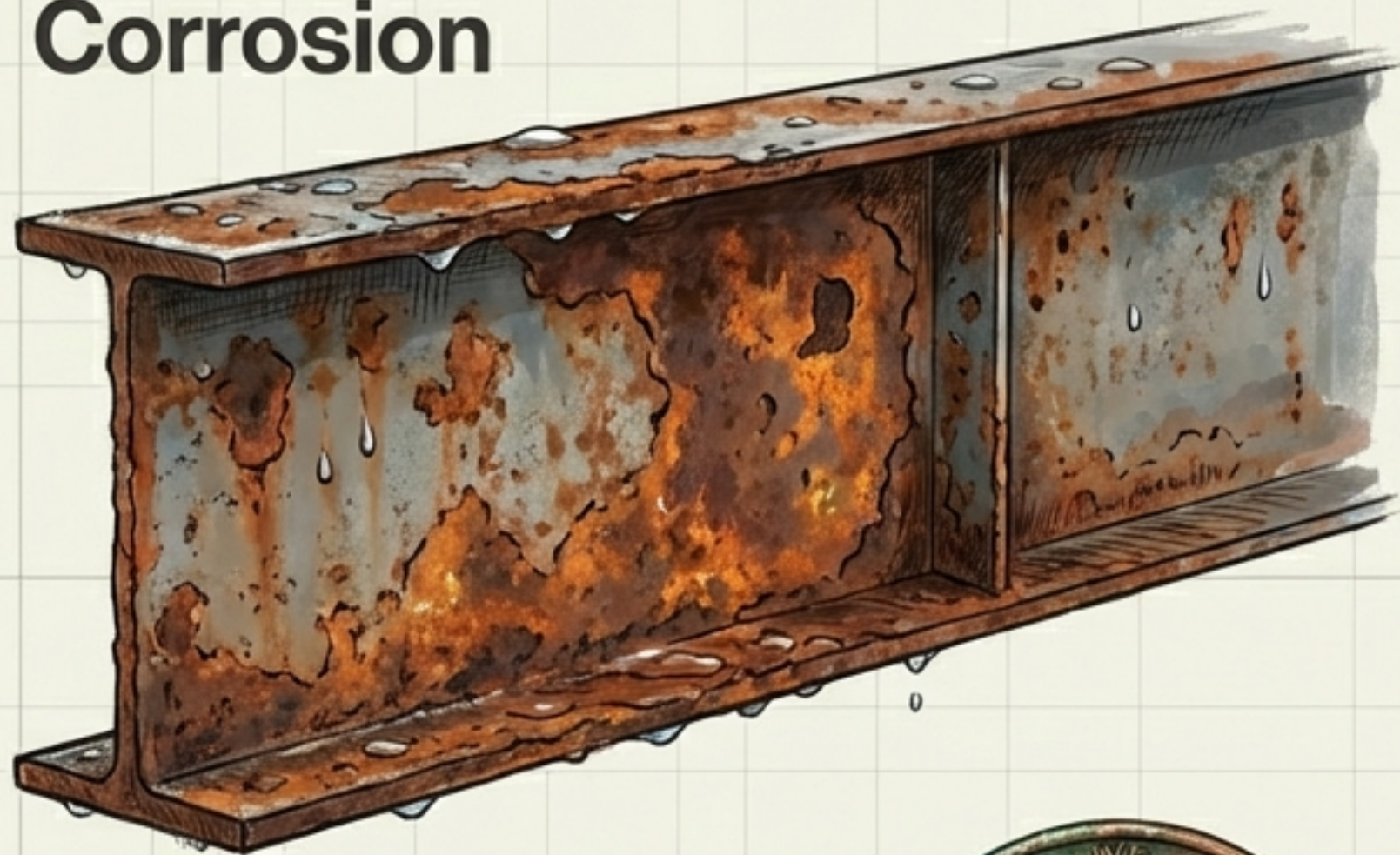
Reduction

Losing Oxygen or Gaining Hydrogen.



The Cost of Oxidation: Corrosion & Rancidity

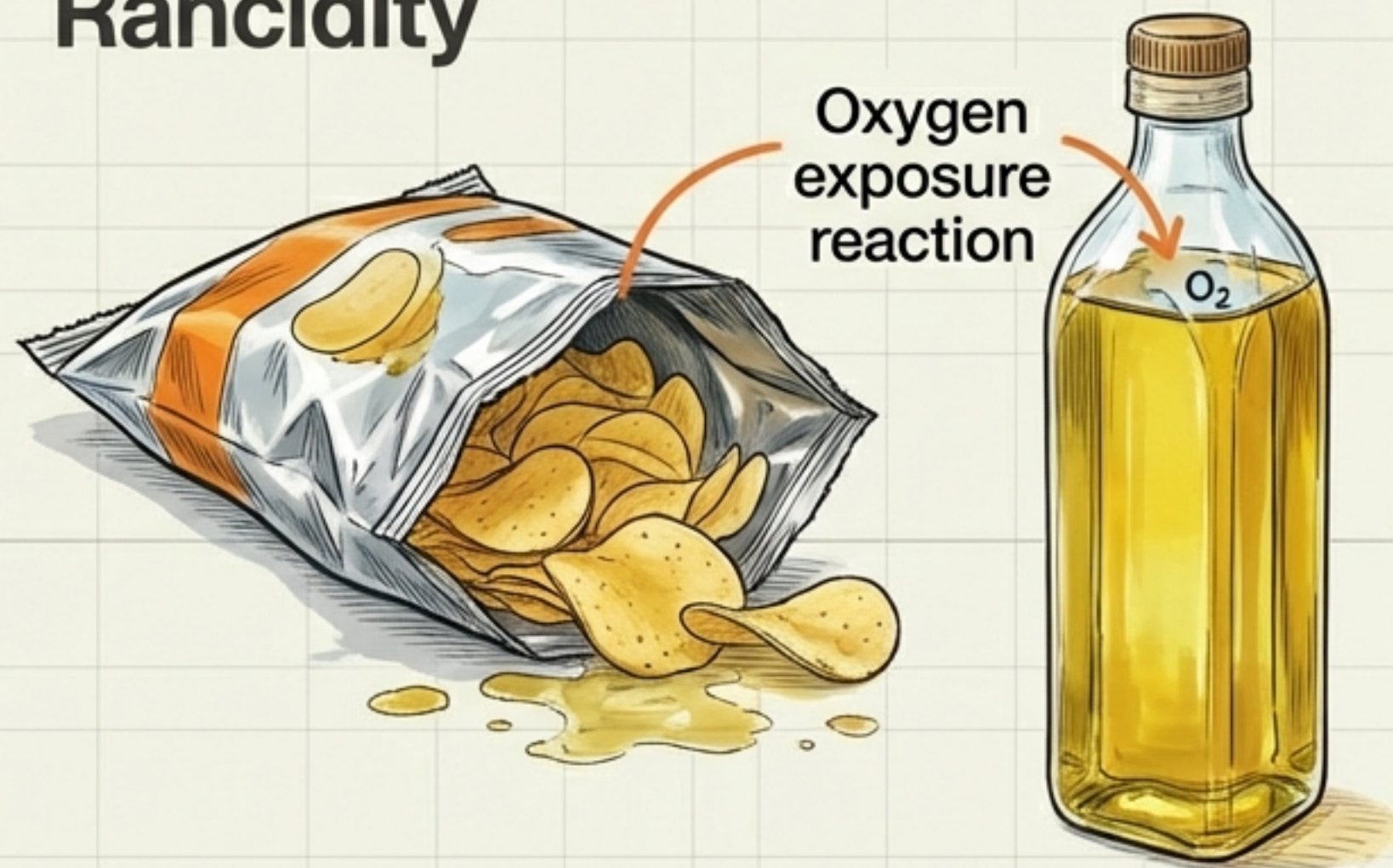
Corrosion



Corrosion: Attack by moisture/acid. Rusting of iron causes massive infrastructure damage.



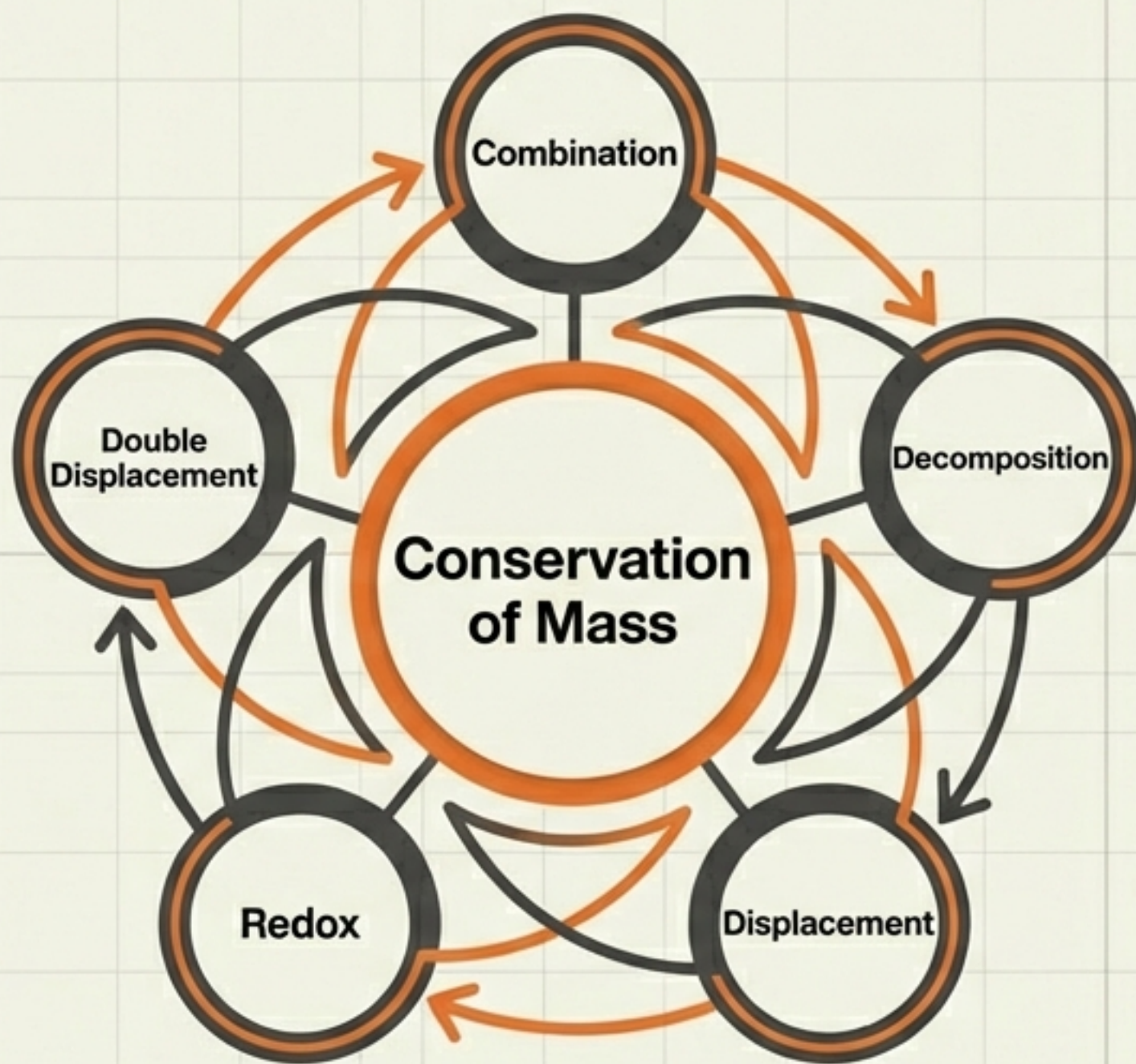
Rancidity



Rancidity: Oxidation of fats/oils causing bad smell and taste.

Solution: Antioxidants and Nitrogen flushing.

The Cycle of Matter



- Key Takeaways

1. Atoms are never lost, only rearranged.
2. Equations are the code we use to balance this reality.
3. From respiration to rust, these reactions drive our world.

“In a chemical reaction, bonds break and new bonds form. The universe is in constant conversation with itself.”